MediLLaMA: AI-Driven Simplification and Multilingual Translation of Medical Reports Using LLaMA-3 and NLLB-200

Mrinal Ashwith S P
Department of Artificial Intelligence and Machine Learning
Sri Ramachandra Institute of Higher Education and Research
Chennai 600 116, Tamil Nadu, India
mrinalashwith@gmail.com
ORCID: 0009-0002-6400-5559

Yuvan Kalyan U
Department of Artificial Intelligence and Machine Learning
Sri Ramachandra Institute of Higher Education and Research
Chennai 600 116, TamilNadu, India
yuvanudayakumar@gmail.com
ORCID: 0009-0005-4730-5877

Hariharan P
Department of Artificial Intelligence and Machine Learning
Sri Ramachandra Institute of Higher Education and Research
Chennai 600 116, Tamil Nadu, India
hari2004.09@gmail.com
ORCID: 0009-0001-8320-066X

Dr. Suja Golden Shiny S
Department of Artificial Intelligence and Data Analytics
Sri Ramachandra Institute of Higher Education and Research
Chennai 600 116, Tamil Nadu, India
sujashiny@sret.edu.in
ORCID: 0000-0002-4145-2084

***Abstract*—Medical reports contain complex clinical terminology, unstructured formats, and specialty abbreviations that hinder patient comprehension, particularly among individuals with limited health literacy or non-English backgrounds. This communication gap leads to misunderstandings, poor treatment adherence, and patient anxiety. As a solution to this problem, MediLLaMA, an end-to-end framework that automatically simplifies and translates clinical text into accessible, patient-friendly language, is proposed. The system integrates BERT -base- Named Entity Recognition (NER) to preserve critical medical terms, LLaMA-3 8B for semantic simplification, and NLLB-200 for high-quality multilingual translation across Indic languages. To achieve a balanced clinical interpretation and to maintain light-weight on the local platform, MediLLaMA uses context-based processing to maintain accurate clinical interpretation with computationally intensive LLaMA-3 component assessed using a cloud-based inference API. Experimental results demonstrate that MediLLaMA significantly improves the ROUGE L, METEOR and BLEU score, when compared to state-of-the-art approaches such as BERTSUMEXT, ACLS and LLaMA3-based MediGen.**

***Index Terms*—Medical report simplification, Clinical text translation, LLaMA-3, NLLB-200**

I. Introduction

Modern healthcare is increasingly encouraging patient involvement; however, complicated medical records continue to be the main barrier to direct involvement. Discharge summaries, clinical notes, and diagnostic results may include domain-specific jargon, and abbreviations that laymen will not easily understand. Such a task is further complicated when dealing with the population in non-English speaking areas like India, where language is a major barrier to comprehension. With the advent of electronic health records and telemedicine, the availability of such reports has improved, but interpretability has remained problematic [13], [14]. This interpretability has become a major challenge for patients to follow post-treatment instructions, resulting in the wrongful use of medications and unfounded anxiety. Considering this interpretability issue, this study presents an end-to-end, AI-based process to ease and render the medical reports into patient-friendly, readable language with the help of advanced Natural Language Processing (NLP) and translation models.

While many of the studies proposed solutions for clinical text summarization to decrease the workload of clinicians and nurses [2] [5] , very few works concentrated on delivering summarized text to the patient for better understanding. Moreover, most of the existing works rely on deep learning models with a sequence-to-sequence encoder-decoder framework [13], [15]. The encoder transforms the medical report into a semantic vector, and the decoder generates the summary. However, these approaches struggle to capture deep semantic features from the input due to the complexity of medical terms and the limitations of shallow network architectures.

Large language models (LLMs) [11] [12], [7] have become a promising solution to learn deep contextual representation due to their architecture, which stacks attention networks. Moreover, studies on large language models determine that the LLaMA-3 model outperforms GPT4 when fine-tuned. Therefore, we adopted to implement and fine-tune the LLaMA-3 model for text summarization and the NLLB-200 model for translating the summarized text to Indic languages. The novelty of the paper lies in developing a stacked model with BERT-Base NER for entity extraction, a LLaMA-3 for text summarization and a medical-aware NLLB-200 for translation.

The major contributions to this paper are as follows:

i) A preprocessing module that normalizes the unstructured N2C2 medical dataset after cleaning, lemmatizing and tokenising the text and then extracts entities to recognise significant medical terms is proposed.

ii) A stacked model that combines LLaMA-3 and NLLB-200 to summarize and translate medical text to Indic languages is proposed.

iii) Hyperparameter tuning is done on NLLB-200 to optimize the performance of the model by systematic variation of parameters such as learning rate, batch size, and attention head.

iv) To further substantiate the translation accuracy, fluency, and entity preservation of MediLLaMA, we compare it with three other baseline models and show that MediLLaMA outperforms them in ROUGE metric and loss values,while maintaining the right balance of efficiency and linguistic fidelity.

The remaining of the paper is structured as follows. Section 2 presents a comprehensive survey of the literature review related to the work based on summarization and translation. Section 3 presents the methodology of MediLLaMA including dataset description, preprocessing and LLaMA-3 and NLLB-200 models. Section 4 presents the experimental results along with experimental setup and evaluation metrics and the section 5 summarizes the conculsion of the work.

II. Related Work

A. Clinical Report Summarization

Recent advances in clinical text summarization have leveraged transformer-based architectures to reduce documentation burden. Ranjan et al. [3] proposed an automatic labelling model that extracts remedies of unstructured medical blog data using NLP modules, such as formal term extraction and LDA-based topic modelling, and POS-based syntactic pattern parsing.

Diomaiuta et al. [1] proposed an automated clinical summary system that was developed to extract patient problem lists from structured and unstructured Italian medical documents. The NLP pipeline integrates language identification, tokenization, POS tagging, parsing rules, and medical entity recognition. The ClinicalBert model was fine-tuned to summarise the report. Vinod et al. [2] optimized the BERTSUM extractive summarizer on clinical reports, which enhanced readability without any improvement in evidence selection. As it is an extraction method, it has no re-writing/ simplification of jargon-laden text, hence restricting the understanding of the patient.

Jiajun Zhang et al. [7] presented an Abstractive Cross-Language Summarization (ACLS) framework, which translates source documents and optimizes the generation of the summaries with the help of Predicate-Argument Structures (PAS). Singhal et al. [18] presented Med-PaLM 2, an answerer of clinical questions based on PaLM-2. The model applies medical domain fine-tuning and sophisticated prompting techniques that include ensemble refinement and chain-of-retrieval. Lee et al. [19] described BioBERT, a biomedical language representation model obtained by pre-training BERT on PubMed abstracts and on PMC full-text articles. The model enhances the comprehension of biomedical terms and setting.

Nori et al. [20] benchmarked GPT-4 on medical challenge problems such as USMLE and MultiMedQA. The research indicated that GPT-4 was correct on questions in USMLE-style more than 86% and was better than prior versions, including GPT-3.5. This finding shows the promise of large language models in medical reasoning and health care use.

B. Clinical Report Translation

Selvararajan et al. [10] created a medical translation system based on AI and the Afrikaans-English language, with disease prediction. The system involves a GRU-based encoder-decoder NMT model that is trained on 15,203 pairs of sentences and is evaluated by BLEU scores.

Nahiyan et al. [4] presented a scientific text classification model named SciTexC based on mBERT, XLM-RoBERTa that operates on Bengali. It also demonstrated the efficiency of multilingual transformers in low-resource languages. Results highlighted the importance of dataset quality and preprocessing in specialized NLP domains.

The work proposed by Zhai et al. [8] suggested the automated system of Neural Machine Translation of medical and pharmaceutical texts in multiple languages. The hybrid model is a Transformer-based M2M-100 model refined with Golden Jackal Optimization to work with complicated medical language and semantic peculiarities.

Surve et al. [9] proposed a more advanced translation system to convert the English medical text reports to the Marathi language by solving the problem of transliteration. The hybrid model is based on initial translation with the help of Google Translate and Microsoft Bing Translator, followed by a modification of the results with the help of a custom medical dictionary that substitutes the translated medical terminology through substitute Marathi translations.

From the above studies, it is clear that most of the work concentrated on the NLP pipeline and abstractive text summarisation with the sequence-to-sequence encoder-decoder frameworks. Very few papers concentrated on using transformer architecture for text summarization, but they are not domain-specific. Furthermore, most of the models either perform summarization or translation exclusively, and not both. To this end, we have proposed MediLLaMA, which performs both summarization and translation of medical records, making use of large language models.

III. METHODOLOGY

A. Dataset Description

In this study, the N2C2 (National NLP Clinical Challenges) [16] [17] dataset was only taken as an input source to test the text-processing pipeline proposed. The data includes de-identified clinical discharge summaries with the following sections: medical history, diagnoses, treatments, and follow-up instructions. The end-to-end workflow of the system was tested using these documents that contain domain-specific terms and abbreviations. In particular, the N2C2 notes were used as real-life clinical contributions to the assessment of preprocessing, named entity recognition, LLaMA-3-based summarization, and NLLB-based multilingual translation. N2C2 data did not undergo the model-training phase, and this aspect guaranteed that the pipeline formed a generalizable system of inference that works with unseen clinical text.

B. Model Architecture

The proposed architecture integrates many NLP components for efficient text medical summarization and translation. The workflow begins with the preprocessing of the N2C2 dataset, which involves section segmentation, text cleaning, such as removing stop words, normalization, and tokenization. NER is used to identify key medical terms, thereby preserving medical accuracy. The processed data is then passed through the LLaMA-3 8B model for contextual understanding and semantic enhancement. Then, the NLLB-200 model generates simplified and translated outputs, in multiple languages, providing readability while retaining medical integrity. This hybrid pipeline ensures that medical text remains both technically correct and accessible to readers who are non-specialists in this domain. Overall, the architecture bridges the gap between clinical documentation and patient comprehension by combining deep learning and NLP with linguistic precision Fig. 1 illustrates the overall model architecture.

C. Preprocessing and NER

The first stage involves text preprocessing, where the raw clinical narrative is normalized through tokenization, removal of stopwords, and lemmatization to reduce lexical variability. Formally, if $D = d_1, d_2, \ldots, d_n$ represents a document corpus. Preprocessing transforms each document d_i into a normalized sequence such that unnecessary tokens are filtered, and words are reduced to their lemma form, ensuring consistent input for downstream models. The next step applies NER using a BERT-based NER model fine-tuned and trained on biomedical corpora. For an input sequence of tokens $x = x_1, x_2, \ldots, x_n$, the encoder produces contextual embeddings $m_i = BERT(x_i)$ where $m_i \in R^{768}$, which are then classified into entity labels z_i via a softmax function:

$$P(z_i|x) = softmax(Wm_i + b), \tag{1}$$

where W and b denote the model parameters. This ensures that crucial medical concepts such as drug names, conditions, and procedures are accurately extracted and preserved.

Fig. 2 shows the preprocessing pipeline employed in MediLLaMA whereby raw clinical text is first partitioned into meaningful units and cleaned of noise, including repeated headers, OCR artefacts and non-informative patterns. The purified text is then normalized by lowercasing, removal of stopwords, alphanumeric filtering, abbreviation processing and lemmatization such that all the inputs are in a uniform linguistic format. The text is then processed with the relevant subword tokenizer, WordPiece, in the case of BERT-based NER extraction. This homogeneous normalization integrated with the encoding pipeline guarantees stable and entity-conserving representations, which is far much more advantageous than only the success of NER and the trustworthiness of downstream LLaMA-3 simplification and multilingual translation.

D. LLaMA-3 8B

The simplified entity-aware text is then passed into LLaMA3 [11] , a large-scale language model, for semantic simplification. LLaMA-3 8B generates text by estimating the conditional probability of each token given its context, defined as:

$$P(y|x) = \prod_{i=1}^{T} P(y_i|y_{<i}, x; \theta) \tag{2}$$

where θ represents trainable parameters, x denotes the input tokens, y denotes the output phrase, $y_{<t}$, denotes all tokens preceding time step t, y_t is the output phrase at timestep t, and T is the length of sequence. This mechanism allows the model to rewrite complex clinical phrases (e.g., "acute myocardial infarction") into simplified equivalents ("heart attack"), while preserving medical accuracy.

To improve the semantic coherence of generated summaries, LLaMA-3 8B is used in inference mode, and its multi-head self-attention is exploited to model long-range dependencies within the clinical text. The architecture described by the model, including RMSNorm, grouped-query attention and SwiGLU feed-forward layers enables efficient processing of medical text with stability and contextual accuracy. The model is not fine-tuned; instead, LLaMA-3 is utilized as a pre-trained instruction-aligned model accessible via remote inference and generates simplified clinical descriptions from a learned distribution. Given an input clinical sequence x the model produces a simplified sequence *P(y)* most likely simplified continuation under its pre-trained parameterization ϕ. This transformation can be formulated as,

$$P(y) = \text{LLaMA-3}(x; \theta), \tag{3}$$

Since no supervised parameter updates are performed, the model relies entirely on its pre-trained capability to simplify and restructure complex clinical text. This approach avoids computational overhead while benefiting from LLaMA-3's strong language modeling and instruction-following abilities, enabling it to outperform earlier architectures such as mT5, BART, or BioBERT in terms of fluency and contextual alignment during simplification. The simplified output $P(y)$ is now

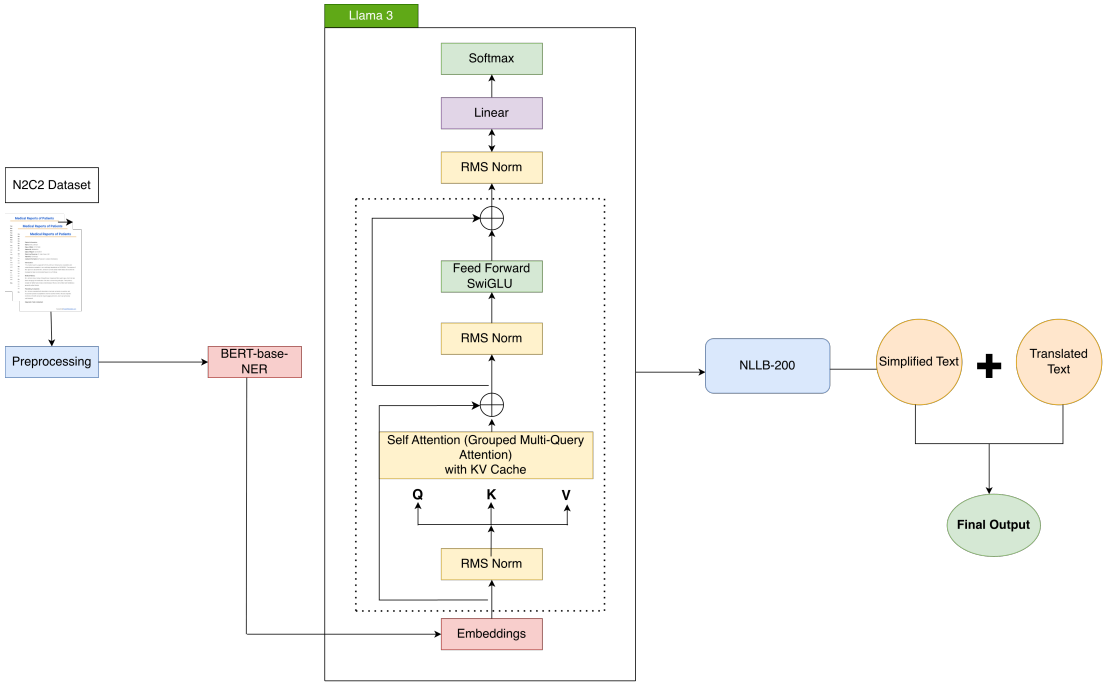


Fig. 1: MediLLaMA Framework for Medical Report Summarization and Multilingual Translation

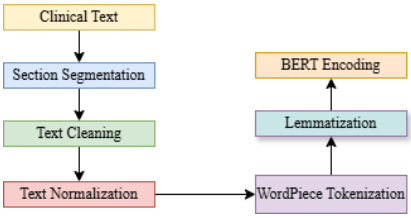


Fig. 2: Preprocessing Pipeline for MediLLaMA Framework

further summarized using the equation 4 and the summarized output $\hat{S}$ is obtained.

$$\hat{S} = F_{\phi}(P(y)), \tag{4}$$

where $F_{\phi}(\cdot)$ denotes the LLaMA-3 inference function. LLaMA-3 8B in this framework functions exclusively as a forward-pass generative model, using its pretrained weights to interpret and simplify complex clinical narratives. The model processes the input sequence autoregressively, estimating the next-token distribution based on previously generated context, which results in a coherent and medically consistent simplification. The transformation $\hat{S} = F\phi(P(y))$, arises from the model's internal architecture—its deep attention stacks, RMS-normalized layers, and SwiGLU-driven feed-forward pathways—which collectively enable precise handling of long, jargon-heavy medical text. Because the model is deployed without any task-specific adaptation, the simplification quality stems directly from its large-scale pretraining and instruction alignment, making it well-suited for producing fluent summaries even in zero-shot settings.

E. Fine Tuning NLLB-200

The simplified English summaries are subsequently processed through the NLLB-200 [6] model for multilingual translation. The NLLB-200 model is a multilingual transformer designed for translation across 200 languages, using an encoder-decoder setup and a shared SentencePiece subword vocabulary to handle morphologically rich, low-resource languages such as Tamil. It features multiple encoder and decoder layers with multi-head attention, feedforward networks, residual connections, layer normalization, and positional encodings to capture long-range dependencies and maintain sequence order. The NLLB-200 model was further fine-tuned with LoRA using the MediTranslate dataset to improve the use of clinical terminology and accuracy of translation when applied to healthcare reports. Furthermore, to enhance the quality of translation of clinical text, it is fine-tuned according to the hyperparameters summarized in Table 1. As early as the fine-tuning stage, though, the very low learning rate 2×10^{-5} and the batch size of 32 provided stable optimization results while 10 training epochs allowed the fine-tuned model to retain the gradual adaptation to the domain-specific linguistic characteristics of the data. Linear warm-up and weight-decay schedules were employed to prevent overfitting, and a maximum sequence length of 512 tokens was utilized to fine-tune the model on the long medical sentences. Following training, which was carried out by converting the English clinical summaries (eng_Latn) into their target-language versions (tam_Taml), the translations became much more accurate and context-preserving.

NLLB operates within a sequence-to-sequence architecture, where the encoder maps the summarized text $\hat{S}$ into hidden

representations h, and the decoder generates the translated sequence $\hat{T}$. Let $\hat{S} = (\hat{s_1}, \hat{s_2} \ldots \hat{s_n})$ be a summarized sentence and $\hat{T} = (\hat{t_1}, \hat{t_2}, \ldots \hat{t_m})$ be the target translated sentence. The NLLB models the conditional probability of the target given the source with a Transformer-MoE network parameterized by Φ.

$$P_\Phi(\hat{T}|\hat{S}) = \prod_{m=1}^{M} P_\Phi(\hat{t}_m|\hat{t}_{<m}, \hat{S}) \tag{5}$$

Training finds parameters given in equation (6) that minimizes a sum of two terms over the dataset D: a sequence cross-entropy term and an auxiliary MoE balancing term.

$$\Phi^* = argmin_\Phi \sum \left[L_{CE}(\hat{T}, S; \Phi) + \lambda L_{MoE}(\hat{S}; \Phi) \right] \tag{6}$$

Where L_{CE} represents the negative log-likelihood of the correct translation, and L_{MoE} encourages the gating network route tokens roughly evenly across experts; λ controls the strength of the regularization.

TABLE I: Hyperparameters fine-tuned in the NLLB-200 model

Hyperparameter	**Value**
Learning Rate	2×10^{-5}
Batch Size	32
Number of Epochs	10
Warmup Steps	500
Weight Decay	0.01
Save Steps	500
Evaluation Steps	500
Maximum Sequence Length	512
Source Language Token	eng_Latn
Target Language Token	tam_Taml

IV. Experimental Results

In this section, we present the experimental results, along with experimental setup, baseline models for comparison and the evaluation metrics.

A. Experimental Setup

Remote inference was used on all experiments involving summarization, with OpenRouter API, which offers hosted access to LLaMA-3 models of Meta on high-performance GPU servers. A request generated about 700-1200 input tokens and 300-600 output tokens, which was not even close to the 8000 token sequence maximum allowed by the service. The use of API was based on a token-based pricing of OpenRouter, and the input and output tokens were added to the cost of computation. Local training or fine-tuning was not done, and all the summarization was created completely by remote execution on the GPU.

Our approach is compared with baseline frameworks as shown below:

(i) BERTSUMEXT developed by Vinod et. al [2], gives extractive summaries, but can not simplify jargon or work with multilingual results. Our model addresses this by doing entity-sensitive simplification and generation of patient friendly translations.

(ii) LLaMA3-based MediGen by Leong et. al [5], does not have medical-term disambiguation and support in multiple languages. We bridge this gap by our approach of NER-based simplification and translation to Indic-language.

(iii) The ACLS framework developed by Zhang et al [7], facilitates the process of cross-language summarization, yet it uses rule-based PAS fusion and fails to deal with clinical terminology. This is enhanced in our model by employing stacked model to provide fully neural and accurate medical simplification and translation.

B. Evaluation Metrics

1) ROUGE:: It compares the quality of text by checking n-gram overlap on text generated on the one hand and reference text on the other. It pays attention to recall the extent to which the reference content has been captured. The typical variants are ROUGE-L and ROUGE-N.

$$ROUGE\text{-}N = \frac{\sum_{S \in Ref} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in Ref} \sum_{gram_n \in S} Count(gram_n)} \tag{7}$$

2) BLEU:: It evaluates the accuracy of matching n-grams between system-generated and reference texts, but NBA undervalues the brevity of short results using a brevity penalty (BP).

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n \right), \tag{8}$$

$$BP = \begin{cases} 1 & \text{if } a > r \\ e^{(1-\frac{r}{a})} & \text{if } a \leq r \end{cases} \tag{9}$$

and where p_n is n-gram precision, w_n is weight for each n-gram, a is candidate length and r is reference length. Greater BLEU translations are more accurate and easier to understand.

3) METEOR:: It uses precision, unigrams recall, and unigrams alignment to determine the quality of the translation. It refers to stemming, synonyms and paraphrases.

$$F_{mean} = \frac{10PR}{R + 9P} \tag{10}$$

$$METEOR = F_{mean} \times (1 - Penalty) \tag{11}$$

$$Penalty = 0.5 \times \left(\frac{chunks}{matches} \right)^3 \tag{12}$$

P and R are the unigram recall and precision. Chunks is the number of contiguous matched word sequences. The penalty minimizes matches with fragments so as to provide semantic consistency and fluency in grammar.

4) Length Ratio:: It measures the proportionate change in the length of generated and reference sentences. It makes sure the simplified or translated outputs are not too brief or too long.

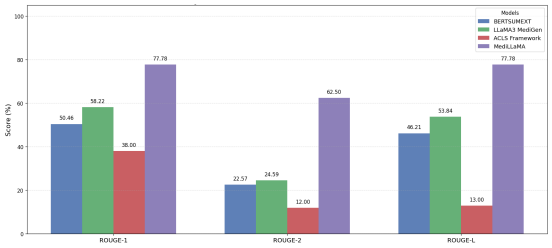


Fig. 3: ROUGE score comparison between MediLLAMA and baseline models

5) Character Length Ratio:: It is used to examine both generated texts and reference texts by the number of characters involved, and testing their compactness and conciseness. This measure is highly applicable to estimate the compression obtained during the text simplification task while preserving readability and major medical terms.

6) Ratio Character Overlap Scores:: The Overlap Ratio indicates how many common characters there are in a system-generated and a reference text, representing superficial and recalling similarities and terminology. A greater overlap ratio implies that key textual features, e.g., named entities or terminology, have been properly maintained by simplification and translation.

TABLE II: Scores for the evaluation metrics

Metric	Score
ROUGE-1	0.7778
ROUGE-2	0.6250
ROUGE-L	0.7778
BLEU	0.59
METEOR	0.3908
Length Ratio	1.2333
Character Length Ratio	1.0942
Character Overlap Score	0.8913

C. Result Analysis:

The fine-tuned version of NLLB shows an evident trade-off between semantic accuracy and translational fluency, and its usefulness in extracting medical information. It scored high in ROUGE-1 and ROUGE-L with 0.7778 and a character overlap of 0.8913, which proves that it preserved the important medical information and the general text structure. The relatively low BLEU 0.59 and medium METEOR 0.3908 scores are, however, due to the contextual rephrasing characteristic of the model, in which the model places more importance on meaning and domain relevance as opposed to exact lexical matching. The Length Ratio 1.094 also shows the presence of compact outputs that are rather fact-oriented and also stress the necessary information rather than wordy sentence constructions.

Fig. 3 shows the performance of ROUGE-1, ROUGE-2, and ROUGE-L of MediLLAMA in comparison with three baseline models. In all ROUGE measures, there is a clear and consistent improvement of MediLLAMA compared to

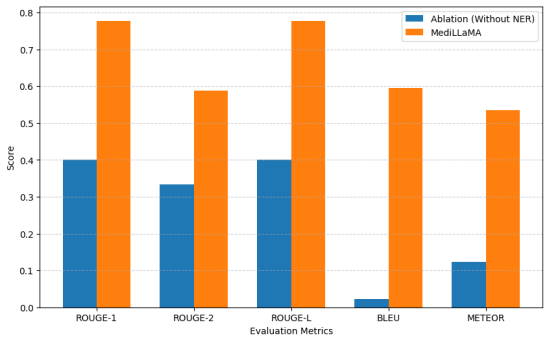


Fig. 4: Performance Comparison between MediLLaMA and Ablation-without NER

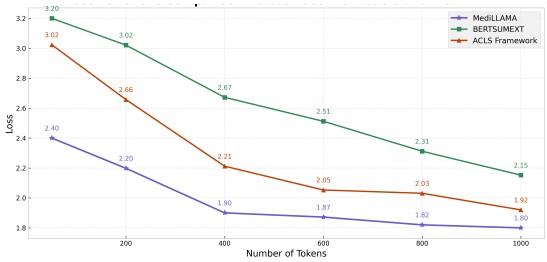
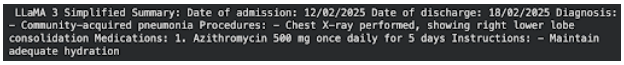


Fig. 5: Loss comparison between MediLLAMA and baseline models

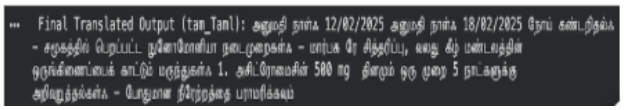
competing methods. It has the largest ROUGE-1 and ROUGE-L scores of 77.78%, which means it has good lexical overlap as well as content preservation. MediLLAMA has much better results in ROUGE-2 (62.50%), pointing to a higher level of coherence and consistency of the bigram level. Conversely, the ACLS structure demonstrates the lowest performance in all measures, and BERTSUMEXT and LLaMa3-MediGen do not provide high performance yet are significantly lower than MediLLAMA. Fig. 4 shows the performance comparison of MediLLaMA (with NER) and the ablation model (without NER). It is clear from the results that when BERT-base-NER is used, the ROUGE-1, ROUGE-2, and ROUGE-L scores are better than the model without NER.

Fig. 5 shows that the loss of the model decreases with the increase in length of tokens in MediLLAMA as well as in two baseline systems. All the models exhibit a decreasing trend of losses as the number of tokens increases, which means that long input sequences are more stable. Nevertheless, the lowest loss is always in MediLLAMA regardless of the token size, which is more efficient in optimization and can better generalize even in longer sequences. However, BERTSUMEXT and ACLS show significantly greater loss at all times, which indicates that MediLLAMA has a greater ability to process clinical text with fewer errors.

Fig. 6a shows the output screenshot of LLaMA-3. The model captures the important information, as shown in the Fig. 5a, including the date of admission and the time of discharge, the diagnosis of the community-acquired pneumonia, and the findings of the chest X-ray of the right lower-lobe



(a) Output screenshot of text summarization using LLaMA-3 model



(b) Output screenshot of text translation using NLLB-200 model

Fig. 6: Model outputs of summarized text and translated text

consolidation. The prescription of medication, Azithromycin 500 mg daily in one dose for five days, is also summarized in a manner that is easy to follow. Lastly, it entails necessary discharge orders, which should advise the patient to ensure he/she is hydrated to recover.

Fig. 6b shows the output of the translated message generated by the NLLB-200 model. The Tamil translation generated by the model correctly includes all important clinical information, such as admission and discharge dates, community-acquired pneumonia diagnosis and chest X-ray results of right lower-lobe consolidation, and correctly translates medication instruction - Azithromycin 500 mg once daily for five days - into well-understood Tamil with no ambiguity, and simple discharge instructions for adequate hydration in patients' language.

V. Conclusion

This paper introduces a stacked model named MediLLaMA, an integrated clinical NLP system that fills one of the most enduring voids in healthcare, simplifying and multilingualizing dense medical documentation into an easy-to-understand text. The system offers a unified pipeline consisting of BERT-based NER to preserve terminologies, LLaMA-3 8B to execute high-quality semantic simplification, and NLLB-200 to execute faithful multilingual translation to ensure a high degree of clinical accuracy and readability. Experimental trials on the N2C2 dataset indicate that the ROUGE score is significantly improved compared to strong baselines, including BERT-SUMEXT, MediGen, and ACLS, which prove that content retention and consistency are quite high. Although BLEU scores are indicative of anticipated paraphrasing of generative simplification models, high scores on METEOR and character-overlap justify semantic faithfulness and domain accuracy.

References

[1] C. Diomaiuta, M. Mercorella, M. Ciampi, and G. De Pietro, "A novel system for the automatic extraction of a patient problem summary," IEEE Symposium on Computers and Communications (ISCC), pp. 182–186, IEEE, July 2017.

[2] P. Vinod, S. Safar, D. Mathew, P. Venugopal, L.M. Joly, and J. George, "Fine-tuning the BERTSUMEXT model for Clinical Report Summarization," International Conference for Emerging Technology (INCET), pp. 1–7, IEEE, 2020.

[3] H. Ranjan, S. Agarwal, A. Prakash, and S.K. Saha, "Automatic labelling of important terms and phrases from medical discussions," Conference on Information and Communication Technology (CICT), pp. 1–5, IEEE, 2017.

[4] F. Nahiyan, and M. M. Hoque, "SciTexC: A Framework for Scientific Text Classification in Bengali using Transformer-Based Techniques," 4th International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), pp. 504–509, IEEE, 2025.

[5] H. Leong, Y. Gao, S. Ji, Y. Zhang, and U. Pamuksuz, "Efficient fine-tuning of large language models for automated medical documentation," 4th International Conference on Digital Society and Intelligent Systems (DSInS), pp. 204–209, IEEE, 2024.

[6] E. Aurelrius, A. H. Sikoko, E. Rakun, and Azizah, K, "Transfer Learning and Custom Loss Applied to Transformer-Based Text Translation for Sign Language Animated Subtitles, " IEEE Access, 2025.

[7] J. Zhang, Y. Zhou, and C. Zong, "Abstractive cross-language summarization via translation model enhanced predicate argument structure fusing," IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 24, no. 10, pp. 1842–1853, 2016.

[8] Y. Zhai, R. Feng and B. Zhao, "Design and Development of Automatic Machine Translation System for Medical and Pharmaceutical Specialties Based on Hybrid Nmt Techniques," International Conference on Intelligent Communication Networks and Computational Techniques (ICICNCT), pp. 01–07, IEEE, 2025.

[9] S.M. Surve, and H. Dand, "Enhancing Translation Quality of English Medical Text Reports into Marathi through integration of Google And Microsoft Bing Transliteration, "OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 4.0, pp. 1–6, IEEE, 2024.

[10] H. Selvarajan, J. Joseph, and S. Roy, "Automated Medical Translation Device: Afrikaans to English with Disease Prediction," International Conference on Artificial Intelligence and Machine Vision (AIMV), pp. 1-6, IEEE, 2025

[11] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey et. al., "The llama 3 herd of models," arXiv preprint arXiv:2407.21783, 2024.

[12] D. Van Veen, C. Van Uden, L. Blankemeier, J.B. Delbrouck et. al., "Adapted large language models can outperform medical experts in clinical text summarization, " Nature medicine, vol. 30, no. 4, pp. 1134–1142, 2024.

[13] A. Phatak, D. W. Savage, R. Ohle, J. Smith, and V. Mago, "Medical text simplification using reinforcement learning (teslea): Deep learning–based text simplification approach," JMIR Medical Informatics, vol. 10, no. 11, e38095, 2022.

[14] Z. Yao, N. S. Kantu, G. Wei, H. Tran et. al., "Readme: Bridging medical jargon and lay understanding for patient education through data-centric nlp," In Findings of the Association for Computational Linguistics: EMNLP, pp. 12609–12629, 2024.

[15] H. Yu, L. Fan, L. Li, J. Zhou et. al.," Large language models in biomedical and health informatics: a review with bibliometric analysis," Journal of healthcare informatics research, vol. 8, no. 4, pp. 658–711, 2024.

[16] S. Henry, K. Buchan, M. Filannino, A. Stubbs, and O. Uzuner, "2018 n2c2 shared task on adverse drug events and medication extraction in electronic health records," Journal of the American Medical Informatics Association, vol. 27, no. 1, pp. 3–12, 2020.

[17] D.Mahajan, J. J. Liang, C. H. Tsou, and Ö. Uzuner," Overview of the 2022 n2c2 shared task on contextualized medication event extraction in clinical notes," Journal of biomedical informatics, vol. 144, 104432, 2023.

[18] Singhal, K., Tu, T., Gottweis, J., Sayres, R., Wulczyn, E., Amin, M., and Natarajan, V., "Toward expert-level medical question answering with large language models. Nature medicine," 31(3), 943-950, 2025.

[19] Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., and Kang, J. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining. Bioinformatics," 36(4), 1234-1240, 2020.

[20] Nori, H., King, N., McKinney, S. M., Carignan, D., and Horvitz, E., "Capabilities of GPT-4 on Medical Challenge Problems," arXiv preprint arXiv:2303.13375, 2023.